Huawei ELEC-ACC-57 Access Node QoS Congestion Event Troubleshooting SOP

Node: ELEC-ACC-57 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_2584

Description

This SOP outlines the steps to troubleshoot and resolve QoS congestion events, specifically microbursts and interface tail drops, on Huawei ELEC-ACC-57 access nodes at the Electronic City Hub. The root cause of these events is typically attributed to misconfigured Quality of Service (QoS) policies, which lead to queue starvation and subsequent congestion. On Huawei equipment, QoS policies are implemented using the Modular QoS Command-Line Interface (MQC). The MQC classifies network traffic into queues based on various parameters, such as source IP address, destination IP address, source port number, destination port number, protocol type, and DSCP value. When QoS policies are misconfigured, traffic may be incorrectly classified, leading to queue starvation and congestion. Furthermore, the Huawei ELEC-ACC-57 access node's architecture, which includes a combination of Gigabit Ethernet interfaces and advanced queuing mechanisms, can exacerbate congestion events. In particular, the node's reliance on hardware-based queuing, such as the Huawei Enhanced Queuing (EQ) mechanism, can lead to interface tail drops when queues become oversubscribed. Additionally, the node's support for advanced QoS features, such as hierarchical scheduling and traffic policing, can further complicate congestion events. Therefore, it is essential to carefully analyze QoS policies and queue configurations to identify and resolve congestion events on Huawei ELEC-ACC-57 access nodes.

Pre-requisite Checks

- Verify that the Electronic City Hub is accessible and that all necessary personnel have been dispatched.

- Check the status of the redundant link to ensure it is operational and not contributing to the congestion event.

- Verify that all QoS policies are correctly configured and not causing queue starvation.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies leading to queue starvation	High
Hardware-based queuing mechanism leading to interface tail drops	Medium
Advanced QoS features not properly configured	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue-statistics, Expected Output: Queue
statistics for all interfaces, Description: This command displays the
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current queue statistics for all interfaces on the Huawei ELEC-ACC-57 access node, including the number of packets in each queue and the average queue length.

Command: display qos policy, Expected Output: QoS policy configuration, Description: This command displays the current QoS policy configuration, including the classification criteria and queue mappings.

Command: display eqq statistics, Expected Output: Enhanced Queuing (EQ) statistics, Description: This command displays the current EQ statistics, including the number of packets dropped due to EQ and the average EQ queue length.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: display qos policy, Description: Review the QoS policy configuration to ensure it is correctly classifying traffic and mapping queues. Check for any misconfigured policies that may be contributing to queue starvation.

Step 2: Step: Adjust QoS policy configuration, Command: modify qos policy classify queue , Description: Modify the QoS policy configuration to correctly classify traffic and map queues. This may involve adjusting the classification criteria or queue mappings.

Step 3: Step: Verify EQ configuration, Command: display eqq statistics, Description: Review the EQ configuration to ensure it is correctly configured. Check for any misconfigured EQ settings that may be contributing to interface tail drops.

Step 4: Step: Adjust EQ configuration, Command: modify eqq , Description: Modify the EQ configuration to correctly configure EQ settings. This may involve adjusting the EQ buffer size or queue length.

Post-Check Verification

```
Command: display qos queue-statistics, Expected Output: Queue statistics for all interfaces, Description: Verify that queue statistics have improved after resolving the congestion event.

Command: display qos policy, Expected Output: QoS policy configuration, Description: Verify that the QoS policy configuration has been correctly modified.

Command: display eqq statistics, Expected Output: Enhanced Queuing (EQ) statistics, Description: Verify that EQ statistics have improved after resolving the congestion event.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo modify qos policy classify queue , Description: Restore the original QoS policy configuration in case the modification caused further issues.

- Step: Restore original EQ configuration, Command: undo modify eqq , Description: Restore the original EQ configuration in case the modification caused further issues.